

Auto detection of schemas

Previously, you learned that a schema is a way of describing how a table is organized. Table schemas can be created manually in Google Cloud's user interface, or by using a JSON file with the command line interface. While these options enable you to write the schema yourself, you can use schema auto detection for datasets that are already organized in CSV files, or existing databases. Schema auto-detection enables you to retain the structure of the columns and rows of your existing file in a new table. In this reading, you'll learn more about schema auto detection, and how to use it with your existing datasets.

Schema auto-detection with BigQuery

Using existing data in CSV, JSON, and Google Sheets, you can use auto-detection in Big Query to create schemas, or the organizational framework for a database.

How to use auto-detection

To enable schema auto-detection in BigQuery, follow these steps:

1. Navigate to the "Explorer" panel and select your dataset
 - Expand the "Actions" option and click "Open"
2. Click "Create table"
3. Under "Source," select your desired source using "Create table from"
 - Under "File format" select CSV, JSONL, or Google Sheets
4. On the "Create table" page in the "Destination" section, select the "Dataset name", "Table name", and "Table type"
 - Verify that the "Table type" is "Native table"
5. Under "Schema", check the "Auto detect" option
6. Click "Create table"

When you select schema auto-detection, Big Query scans the first 500 rows of your data as a sample from a random file in the data source to assign data types to each field. The data type of each field is inferred based on the rows with the most fields, so auto-detection is useful as long as there's at least one complete row with values in every column or field. If you have a dataset with many columns, automatically defining the data type for each of the fields using auto-detection will save you time when you organize your data.

Updating schemas

Once your schema is created, you can review a sample of your data table in BigQuery to ensure that the field types have been labeled correctly. If there are errors, you can edit the schema by following these steps:

1. Navigate to the “Explorer” panel and select your dataset
 - Select the table
2. Click “Schema”
3. Click “Edit schema”
4. Revise the Field name, type, or mode as needed
 - You can also add new fields
5. Click “Save”

Key takeaways

Schema auto-detection can be used with data files, like CSV, JSON, and Google Sheets files. Using auto-detection allows BigQuery to identify the field types for your dataset, and develop the schema without manual input. For large, clean datasets, this option can save you time in setting up your data tables for analysis.

Resources for more information

To learn more about setting up auto-detection in BigQuery’s console, and the command line editor, check out this link:

- Read how Google Cloud provides documentation on setting up auto-detection in many different ways: <https://cloud.google.com/bigquery/docs/schema-detect>